Machine Learning: Advanced Vehicle Management Dashboard

This tutorial guides you to building a styled Django dashboard that displays grouped data and charts.

PART 1: Create Virtual Environment

1. Create project folder:

```
mkdir vehicle_ml_dashboard
cd vehicle_ml_dashboard
```

2. Create virtual environment:

```
python -m venv venv
```

3. Activate virtual environment:

```
Windows:
venv\Scripts\activate

Mac/Linux:
source venv/bin/activate
```

Using requirements.txt (Recommended Method)

To generate requirements.txt automatically:

```
pip freeze > requirements.txt
```

Open it and Add below packages in requirements.txt file:

```
django
pandas
plotly
```

Then install all packages at once:

```
pip install -r requirements.txt
```

PART 2: Create Django Project and App

Create project and app:

```
django-admin startproject config .
python manage.py startapp vehicles
```

PART 3: Add 'vehicles' to INSTALLED_APPS in settings.py.

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'vehicles'
]
```

PART 4: Vehicle Dummy Data

In the root directory, create a folder, name it dummy_data and place the CSV file in it

PART 5: Within vehicle app/folder create a file named dashboard.py

```
import plotly.express as px
import plotly.offline as opy
import plotly.graph_objects as go

import pandas as pd

def frequency_table(df):
    # Simple counts
    manufacturer_counts = df['manufacturer'].value_counts().reset_index()
    manufacturer_counts.columns = ['Manufacturer', 'Count']

    # Convert to HTML using the correct method name: .to_html()
    table_html = manufacturer_counts.to_html(
        classes="table table-bordered table-striped table-sm",
        float_format='%.2f',
        justify='center'
    )
    return table_html
```

PART 6: Within vehicle app/folder, add below code in views.py

```
import pandas as pd
from django.shortcuts import render
from .dashboard import frequency_table

def dashboard_view(request):
    queryset = pd.read_csv("dummy_data/vehicles_data_1000.csv")
    df = pd.DataFrame(queryset)

    return render(request, "vehicles/index.html", {
        "frequency_table": frequency_table(df)
    })
```

PART 7: Create Styled Template within vehicle app/folder

templates/vehicles/index.html

Use Bootstrap in dashboard.html for professional styling.

```
<!DOCTYPE html>
<html>

<head>
    <title>Vehicle ML Dashboard</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
    <style>
        body {
            background-color: #f4f6f9;
        }

        .scrollspy-example {
            height: 80vh;
            overflow-y: scroll;
            position: relative;
        }

        .nav-pills .nav-link.active {
            background-color: #0d6efd;
        }
    </style>
</head>

<body data-bs-spy="scroll" data-bs-target="#dashboard-nav" data-bs-offset="100"
tabindex="0">
    <div class="container-fluid mt-4">
        <div class="row">
            <!-- Sidebar Navigation -->
            <div class="col-md-3">
                <div id="dashboard-nav" class="list-group position-sticky top-
0">
                    <a class="list-group-item list-group-item-action"
href="#frequency-table">Frequency Tables (One-way)
                    </a>
                </div>
            </div>

            <!-- Scrollable Content -->
            <div class="col-md-9">
                <div class="scrollspy-example" data-bs-spy="scroll" data-bs-
target="#dashboard-nav"
                    data-bs-smooth-scroll="true" tabindex="0">
                    <h1 class="text-center text-primary mb-4"> Vehicle Machine
Learning Dashboard </h1>
                    <div id="frequency-table" class="card shadow mb-5">
                        <div class="card-header bg-info text-white">
                            Frequency Tables (One-way)
                        </div>
```

```
                        <div class="card-body">
                            {{ frequency_table|safe }}
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.
js"></script>
</body>

</html>
```

PART 8: URL Configuration

Create urls.py within vehicles app/folder and add below code

```
# vehicles/urls.py
from django.urls import path
from .views import dashboard_view

urlpatterns = [
    path("", dashboard_view, name="dashboard"),
]
```

Add below code in urls.py found in config folder

```
# config/urls.py
from django.urls import path, include

urlpatterns = [
    path("", include("vehicles.urls")),
]
```

PART 9: Running the Application

```
python manage.py runserver
```

Visit: http://127.0.0.1:8000/

If you see below messages, run migrations to resolve it

```
You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s):
admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
```